WHY AUTHENTICATION ISN'T ENOUGH FOR AGENTS

Authentication answers a simple question: Who is making this request?

In traditional software systems, that answer is often enough. Once an application or user is authenticated, authorization policies can grant access based on predefined roles or permissions. Because the application's behavior is predictable, those policies can be written confidently with respect to how the system will operate. *Agentic systems break that assumption.*

An agent can use various tools, access different data, and try different strategies each time it attempts to achieve a goal. Knowing who the agent is doesn't reveal what actions it might take next.

For this reason, agent systems cannot depend solely on authentication. Each action proposed by the agent must be assessed against policy in real-time. Authorization acts as the mechanism that limits the agent's behavior, making sure that even dynamically planned actions stay within safe limits.

18.3 A policy-aware agent loop

The interaction shown in figure 18.1 does not occur just once. As an agent works toward a goal, it repeatedly plans, acts, evaluates results, and revises its strategy. Each step may involve invoking tools, accessing data, or triggering workflows.

In traditional applications, authorization checks are placed at specific points in a predefined workflow. With agents, there is no fixed sequence of steps. The path to completing a task emerges dynamically as the agent explores different strategies. This means authorization decisions can no longer be treated as occasional checkpoints. They must be integrated into the agent's ongoing execution loop.

ACME addressed this challenge by embedding policy evaluation directly into the agent's runtime behavior. Rather than allowing the agent to invoke